In [2]:
```python
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
import matplotlib.pyplot as plt

from sklearn import metrics
from sklearn.metrics import confusion_matrix
from mlxtend.plotting import plot_confusion_matrix
```

In [3]:
```python
X, y = load_iris(return_X_y=True)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=0)
```

In [4]:
```python
dtree = DecisionTreeClassifier(criterion="entropy")
```

In [7]:
```python
clf = dtree.fit(X_train, y_train)
y_pred=clf.predict(X_test)
print("Number of mislabeled points out of a total %d points : %d"
      %(X_test.shape[0], (y_test != y_pred).sum()))
```

```
Number of mislabeled points out of a total 45 points : 1
```

In [8]:
```python
y_pred
```

Out[8]:
```
array([2, 1, 0, 2, 0, 2, 0, 1, 1, 1, 2, 1, 1, 1, 1, 0, 1, 1, 0, 0, 2, 1,
       0, 0, 2, 0, 0, 1, 1, 0, 2, 1, 0, 2, 2, 1, 0, 2, 1, 1, 2, 0, 2, 0,
       0])
```

In [9]:
```python
y_test
```

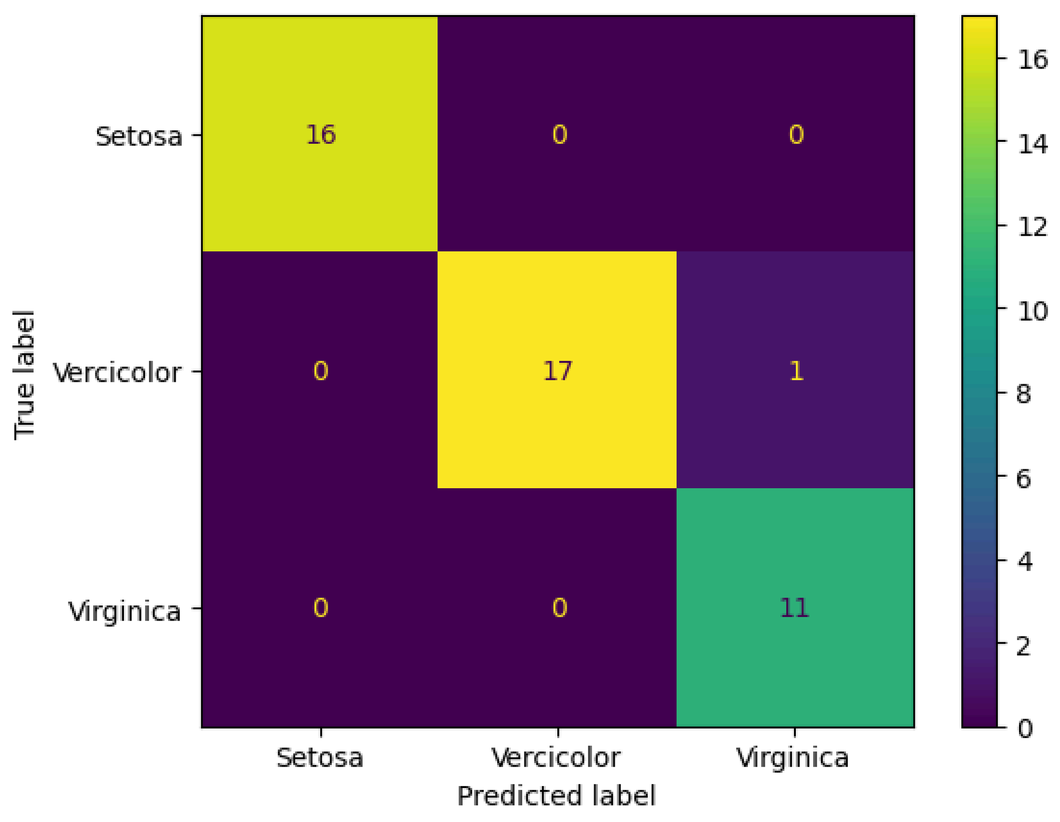
Out[9]:
```
array([2, 1, 0, 2, 0, 2, 0, 1, 1, 1, 2, 1, 1, 1, 1, 0, 1, 1, 0, 0, 2, 1,
       0, 0, 2, 0, 0, 1, 1, 0, 2, 1, 0, 2, 2, 1, 0, 1, 1, 1, 2, 0, 2, 0,
       0])
```

In [12]:
```python
print("Accuracy of the NB Classifie is:" ,metrics.accuracy_score(y_pred, y_test)*100)
```

```
Accuracy of the NB Classifie is: 97.77777777777777
```

In [14]:
```python
cm = confusion_matrix(y_test, y_pred)
cm_display = metrics.ConfusionMatrixDisplay(confusion_matrix = cm, display_labels = ['Setosa', 'Vercicolor', 'Virginica',])

cm_display.plot()
plt.show()
```



In [15]:
```python
res = clf.predict([[5.1,3.5,1.4,0.2]])
if res==0:
    print('Setosa')
if res==1:
    print('Versicolor')
if res==2:
    print('Virginica')
```

```
Setosa
```

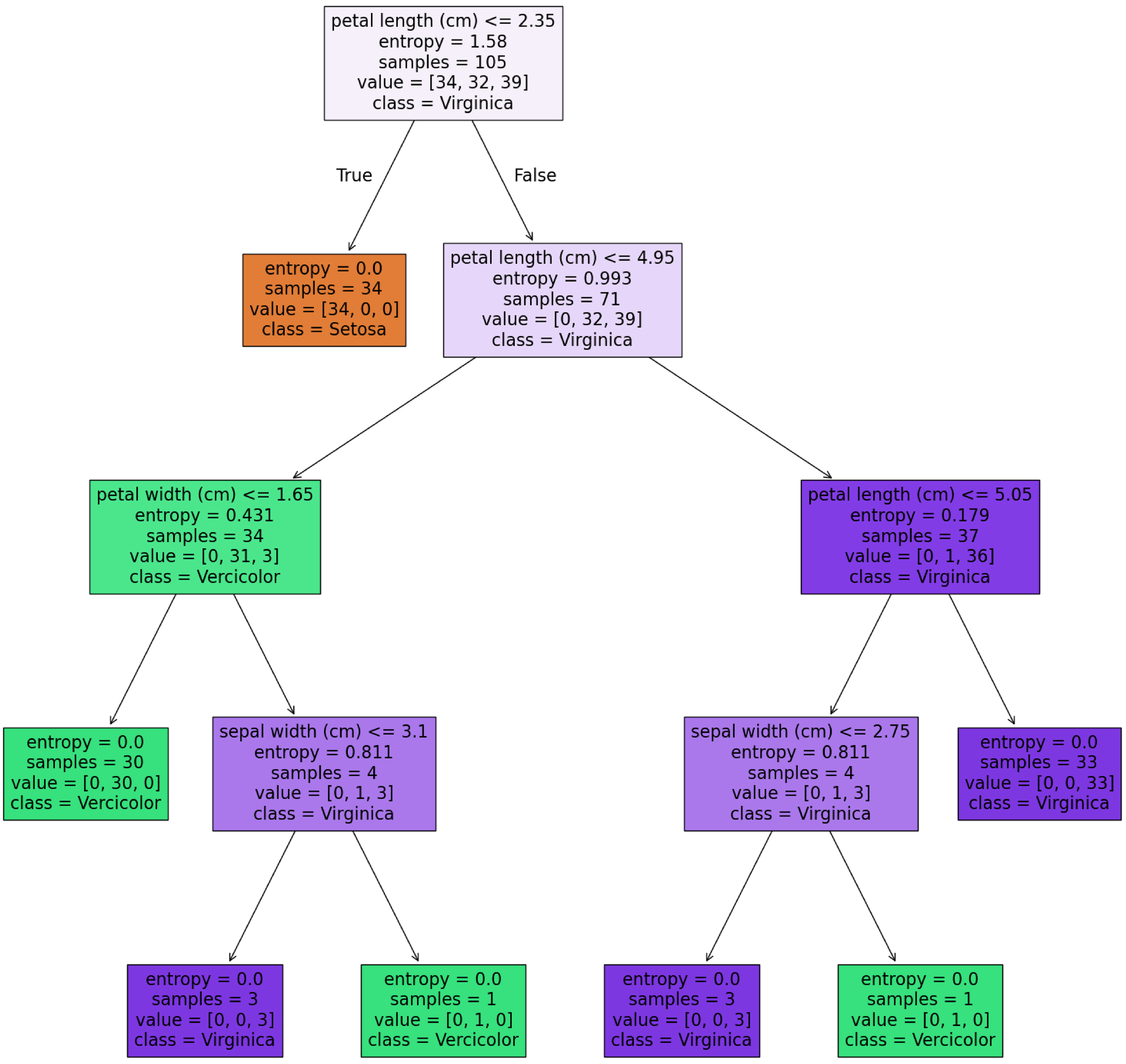
In [16]:
```python
res = clf.predict([[5.8,2.6,4.,1.2]])
if res==0:
    print('Setosa')
if res==1:
    print('Versicolor')
if res==2:
    print('Virginica')
```

```
Versicolor
```

In [17]:
```python
res = clf.predict([[6.8,3.,5.5,2.1]])
if res==0:
    print('Setosa')
if res==1:
    print('Versicolor')
if res==2:
    print('Virginica')
```

```
Virginica
```

In [18]:
```python
from sklearn import tree
fig = plt.figure(figsize=(20,20))
_ = tree.plot_tree(clf,
feature_names=load_iris().feature_names,class_names=['Setosa','Vercicolor','Virginica'],filled=True)
```



In [ ]: